# Bedanta Bandyopadhyay



Electronics Engineer | Software Developer | Full-Stack & Cloud Engineering | ML / Embedded Systems

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## SUMMARY

Software developer building full-stack, cloud-deployed applications, with a B.Tech in Electronics & Communication Engineering (CGPA 8.86) and an active data-structures-and-algorithms practice. Shipped production-style systems spanning React/Node.js applications, PostgreSQL/Redis backends with real-time pub-sub, cascaded LLM/AI integrations, and IoT platforms deployed on AWS, Render, and GitHub Pages. Pairs rigorous problem-solving with a 7+ year record of independently managing client-facing technical projects as a freelance CGI/3D engineer.

## EDUCATION

**Institute of Engineering & Management** *Expected 2027*
*B.Tech, Electronics & Communication Engineering, CGPA: 8.56*

**Ram Mohan Mission High School** *2023*
*Higher Secondary (ISC): 73.2%*

**Ram Mohan Mission High School** *2021*
*Secondary (ICSE): 94.5%*

## TECHNICAL SKILLS

**Languages:** Python, C, Java, JavaScript
**OS Familiarity:** Linux, macOS, Windows
**Web & Backend:** React.js, Node.js, HTML, CSS
**Databases & Real-Time:** PostgreSQL, MongoDB, MySQL
**Cloud, Auth & DevOps:** AWS, Firebase, Docker
**AI / Machine Learning:** Scikit-learn, LangChain, TensorFlow
**Embedded and IoT:** Arduino, Raspberry Pi, Jetson, ESP32
**Simulation:** KiCAD, Cadence Virtuoso, LTSpice, MATLAB
**CAD & 3D:** AutoCAD, Fusion 360, Blender, 3ds Max, Corona
**CG and Post Production:** Adobe Suite, Nuke, DaVinci Resolve

## PROJECTS

**IdleServiceKiller: Multi-Tenant LMS, Coding Judge & Proctored Exam Platform** *Oct '26*
github.com/PandaDsgn/IdleServiceKiller/

- Developed a Windows specific low memory consuming service that runs in the background and frees up CPU and memory allowing more breathing room for active applications. It works specifically on Third-party applications and has a blacklist and whitelist for services and applications.

**Tech:** Rust, Powershell, Bash

**HonorRoll: Multi-Tenant LMS, Coding Judge & Proctored Exam Platform** *Jul '26 – Present*
pandadsgn.github.io/HonorRoll/

- Built a multi-tenant SaaS platform (React web + React Native/Expo mobile) serving four roles (student, teacher, admin, superadmin), combining an LMS, a 10-language sandboxed coding judge (child_process, unprivileged OS user, ulimits, no Docker), and scan-mode grading with deferred OCR.
- Built real-time webcam exam proctoring via on-device computer vision (MediaPipe: face/gaze/phone/speech detection) with browser lockdown, plus cross-org superadmin tooling (audit logs, scoped admin impersonation via org-override headers).

**Tech:** React, React Native/Expo, Node.js, Express, PostgreSQL, S3, MediaPipe, Razorpay

**Infrastructure-Assassin: AI-Driven Infrastructure Testing & Resilience Suite** *Jul '26*

pandadsgn.github.io/Infrastructure-Assassin-Production

- Built a multi-tier, RBAC-governed platform using PostgreSQL for persistent telemetry and Redis for session caching, with LISTEN/NOTIFY pub-sub over Server-Sent Events for real-time dashboard updates.
- Integrated a cascaded LLM pipeline (Gemini API, Ollama, Grok cascaded) for automated threat analysis, added Firebase Authentication, and shipped CI/CD to a multi-instance Render backend + GitHub Pages frontend.

**Tech:** React, Node.js, PostgreSQL, Redis, Firebase, Gemini API, CI/CD, Render, GitHub Pages

**OccuNova: Deep Learning Diagnostic System for Glaucoma Detection** *Jun '26*

pandadsgn.github.io/OccuNova/

- Designed a cascaded ResNet50 computer-vision pipeline with AI-driven UNet cropping to isolate optic-nerve regions from retinal fundus scans.
- Built a scalable inference backend and React/Node.js frontend covering the full lifecycle from clinical data preprocessing to model training and deployment.

**Tech:** Python, CNNs, Computer Vision, Node.js, React, Git/GitHub

**Agro-Guard: IoT Crop & Grain-Silo Monitoring System** *Feb '26*

pandadsgn.github.io/Agro-Guard

- Engineered an Arduino-based sensor pipeline (DHT11, HC-SR04, LDR) for continuous environmental telemetry capture in agricultural storage.
- Built AI-driven analytics for grain storage management on a scalable backend deployed on Render, spanning hardware integration through cloud deployment.

**Tech:** Arduino, IoT sensors, Node.js, React, Render, Git/GitHub

## PROFESSIONAL EXPERIENCE

**3D Generalist & Environment Artist, The AutoFocus Studio** *Aug '25 – Feb '26*

- Delivered high-fidelity visualizations and technical artwork for studio projects and clients across environment design and generalist 3D workflows.
- Worked on the launch campaign for the Hyundai Creta N Line Knight Edition, creating 360° exterior imagery for the website.

**Co-Lead, Ninja Fellowship** *Nov '22 – Mar '25*

- Led a team of 3D artists over 2+ years, coordinating delivery of visual assets for multiple brand engagements.

**Automotive & Product Visualizer (Freelance)** *Apr '19 – Aug '25*

- 7+ years independently managing end-to-end client projects, delivering hyper-accurate 3D design and lighting for automotive and product visualization.
- Collaborated with PJKT on a classified project built on a design by Sasha Selipanov (former Bugatti / Koenigsegg designer).

## INTERNSHIP & CERTIFICATIONS

**Advanced & Cleaner Materials in Solar Technology: Internship at IEM Newtown, IEDC** *Jun '25 – Jul '25*

- Optimized performance of tri-layer, lead-free, carbon-based perovskite solar cells using machine learning; authored a full technical report.

**Certifications:** Business Intelligence & Analytics, NPTEL, IIT Madras (Jun '25) • Cloud Computing, NPTEL, IIT Kharagpur (Jun '25) • Joy of Computing with Python, NPTEL, IIT Ropar (Nov '24) • Introduction to IoT, Coursera, UC San Diego (Oct '24)

## PUBLICATION

**AgroGuard: A Novel IoT Enabled Framework for Smart Crop Preservation in Grain Silos**

*Publisher: Bentham Science Journal | Status: Ongoing*

## LEADERSHIP & EXTRACURRICULAR

- Computer Society of India – UEM: Vice President (Jan '26–Present), Manager (Aug '25–Jan '26), Graphics Lead (Mar '25–Jan '26)
- Iris, UEMK Photography Society: Tech/Media/Graphics Lead (Sep '24–Aug '25)
- Class Representative (Jan '26–Present)